The Empirical Case for Gaussian Renormalization as Default

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Provisional draft, 17 August 2026

Abstract

Given probabilities over mutually exclusive events, and told that one has been *withdrawn*, scratched, delisted or masked out of a softmax, proportional renormalization is the human reflex or, in the case of many models, the architecture. Conditioning is available but not free: it needs a model of how the probabilities arose, and renormalization *tacitly* supplies one. Within the class of independent additive random-utility models it is the model with Gumbel noise; Holman and Marley are credited with that representation [12], and Yellott [18] showed it is not identified from pairwise comparisons alone but is unique once triples are included. Assume Gaussian noise instead and the survivors' odds contract by a computable amount. Both assumptions are calibrated to the same probabilities and fit nothing, so they can be scored against each other. The Gaussian contest forecasts better on an experiment where subjects chose from all thirty-one subsets of five goods, and on twelve of twelve ranking collections. Contraction of a noisy estimate lowers log loss by itself, so we check against populations that renormalize exactly; the advantage survives except in the smallest collection.

1 Two restriction maps

For a population and a menu S, let p_i be the share choosing or first-ranking item i. Luce's axiom [5] assigns a worth $u(i)$ with

$$p_i = \frac{u(i)}{\sum_{j \in S} u(j)}, \tag{1}$$

so the shares determine the worths up to scale. The prediction for a subset $T \subseteq S$ follows by changing the denominator alone, which leaves every ratio p_i/p_j with $i, j \in T$ exactly as it was in S. For a two-item menu the predicted share preferring i is therefore $p_i/(p_i + p_j)$. This invariance is Independence of Irrelevant Alternatives.

The reflex to renormalize is not confined to choice modelling, and it is strong enough to be mistaken for a theorem. Renormalization is the posterior for mutually exclusive events only when the information that one is impossible is uninformative about which of the rest obtained; Monty Hall is the familiar counterexample. Withdrawal is a third case, neither flat-likelihood conditioning nor informative conditioning, because no information about a fixed experiment has arrived: an alternative is gone and the problem is a different one. Conditioning is still available, but it now requires a model of how the probabilities were generated, and renormalization is one such model rather than a consequence of probability theory. Within the class of independent additive random-utility models, Gumbel errors generate Luce choice probabilities, a representation attributed to Holman and Marley [12]; Yellott [18] showed that it is not unique from pairwise comparisons but that Gumbel is uniquely identified once triple comparisons are included. Which generative assumption to make is then an empirical question, and this paper measures two of them.

The independent Gaussian random-utility model, Thurstone's Case V [17] in the paired-comparison case, extends to a multiway contest and connects the same two quantities. Item i carries a location a_i, draws a performance $X_i \sim N(a_i, 1)$ on each occasion, and the highest draw is chosen, so

$$p_i = \int \varphi(x - a_i) \prod_{k \neq i} \Phi(x - a_k)\, dx. \tag{2}$$

The integral is one-dimensional for any number of alternatives, since conditioning on $X_i = x$ makes the other events independent, and it is inverted numerically by a lattice method of [2], which recovers the whole location vector in one pass over a shared grid; locations are identified up to a common translation, fixed here by $\sum_i a_i = 0$. Prediction for a subset T means re-running the contest among T alone. For a pair this is one normal evaluation,

$$P(X_i > X_j) = \Phi\left(\frac{a_i - a_j}{\sqrt{2}}\right), \tag{3}$$

and if $p_i > p_j$ it lies between $\frac{1}{2}$ and $p_i/(p_i + p_j)$: removing the other alternatives contracts the odds toward one. Winning a large field requires an unusually high draw, and beating a single rival does not, so the favourite's advantage counts for less. Proposition 1 makes this exact.

The class of independent additive random-utility models contains exactly two parameter-free points. Fix the noise law up to location and there is nothing left to choose; Gumbel gives renormalization and unit-variance Gaussian gives Case V, and every law between them costs a number. So an analyst holding aggregate shares and no restricted observations has two candidates, not a family, and the comparison below is between the only two defaults available. Anything fitted is a different exercise, and one that cannot be tested by the discrepancy it was fitted to absorb.

Both rules are calibrated to the same $|S| - 1$ free shares, and neither sees any observation from a smaller menu. The comparison is between two maps with no free parameters left, rather than between two fitted models. This also fixes what the comparison can settle. A fitted Plackett–Luce likelihood [14] uses the whole ranking, and a fitted Bradley–Terry model [10] uses the pairwise outcomes directly; neither is estimated here. The question is which map from full-menu shares to restricted-menu shares predicts population choice better. Both are *representative-agent* maps: each treats the population as one chooser with one set of worths or locations, and transports its shares from S to T.

The direction of the disagreement is a theorem, and it holds for noise laws other than the Gaussian. Let f and F be the density and distribution of the common noise, assumed independent and identically distributed across alternatives up to location, with positive density on a common support and $\log r$ differentiable, and let $r = f/F$ be its reverse hazard, with the highest draw winning as above.

Proposition 1 (Removing alternatives contracts the odds)**.** *Suppose* $\log r$ *is concave. Let* $T \subset S$ *and* $i, j \in T$ *with* $a_i > a_j$*. Then*

$$\frac{P(i \mid S)}{P(j \mid S)} \;\geq\; \frac{P(i \mid T)}{P(j \mid T)} \;\geq\; 1,$$

with the first inequality strict when $\log r$ *is strictly concave and* $S \setminus T$ *is non-empty.*

Proof. Write $A(x) = f(x - a_i)F(x - a_j)$ and $B(x) = f(x - a_j)F(x - a_i)$ for the two items in question, $C(x) = \prod_{k \in T \setminus \{i,j\}} F(x - a_k)$ for the other survivors, and $H(x) = \prod_{k \in S \setminus T} F(x - a_k)$ for the items removed. Then $P(i \mid T) \propto \int AC$ and $P(i \mid S) \propto \int ACH$, and likewise for j with B in place of A. Now

$$\frac{A(x)}{B(x)} = \frac{r(x - a_i)}{r(x - a_j)},$$

which is increasing in x when $\log r$ is concave, since $a_i > a_j$ places $x - a_i$ below $x - a_j$ and $(\log r)'$ is decreasing. Under the probability measure proportional to $B(x)C(x)\,dx$ the functions A/B and H are both increasing, hence positively correlated, so

$$\frac{\int ACH}{\int BCH} \geq \frac{\int AC}{\int BC},$$

which is the first inequality, strict under strict concavity when $S \setminus T$ is non-empty. For the second, raising a_i stochastically increases X_i and so raises $P(i \mid T)$ while lowering $P(j \mid T)$, and the two are equal at $a_i = a_j$ by exchangeability; hence $a_i > a_j$ gives $P(i \mid T) \geq P(j \mid T)$. $\square$

The proposition says that removing alternatives moves the ratio toward one.

Two laws sit at the ends of that condition. The Gaussian satisfies it strictly, since $\frac{d^2}{dx^2} \log r(x) = -\operatorname{Var}(Z \mid Z < x) < 0$. The Gumbel has $r(x) = s^{-1}e^{-x/s}$, so $\log r$ is affine, the inequality becomes an equality, and renormalization is recovered exactly. That is the Holman–Marley representation [12]. The axiom is a boundary of the class.

The boundary is where machine learning sits. A softmax over scores z_i is Equation (1) with $u(i) = \exp(z_i)$, so masking a softmax and rescaling the survivors is proportional renormalization, and by Yellott's theorem that is the Gumbel-noise member of the contest class. The default is a single point of the family.

The proposition gives the direction of the effect, not its size. Size depends on the shares as well as on the noise law: recalibrating several laws to one share vector and then to another can reverse which of them contracts more. We therefore make no claim that thinner or heavier tails contract more.

2 Scoreboard

Nineteen comparisons are reported below, each one a population whose full-menu shares calibrate both maps and whose restricted-menu choices score them. A comparison is a win for the race when its gain exceeds the gain a renormalizing population would produce, with a Monte Carlo tail frequency of 0.05 or less, a loss when renormalization exceeds the race on the same test, and a draw otherwise. The null matters because contraction of a noisily estimated share vector lowers log loss on its own, which Section 7 quantifies.

Two features of the table matter more than the count. Every loss column is empty, so on no population examined does renormalization beat the race by this test. And the draws are concentrated where they should be: the smallest ranking collection and the smallest consumer experiment.

[1] See Section 4 for qualifications. The quoted tail is for the frame requiring the most arithmetic; the result does not hold in the frame requiring the least, which Section 4 argues is the thesis working rather than failing.

Population	Restriction	Tail	Verdict
Occupational prestige	ranking-induced	$\leq .005$	win
Sushi	ranking-induced	$\leq .005$	win
Political goals	ranking-induced	$\leq .005$	win
GSS socialization	ranking-induced	$\leq .005$	win
GSS job values	ranking-induced	$\leq .005$	win
Jester files 1, 2, 3	ranking-induced	$\leq .005$	win $\times 3$
Dots	ranking-induced	$\leq .005$	win
Puzzles	ranking-induced	$\leq .005$	win
Netflix	ranking-induced	$\leq .005$	win
Sports participation	ranking-induced	.51	draw
Consumer goods, experiment 1	observed menus	.005	win
Gambles over token prizes	observed menus	.035	win[1]
US state ballots 2024	observed ballots	.005	win
Consumer goods, experiment 2	observed menus	.129	draw

Table 1: Thirteen wins, two draws, no losses. The upper block reads subset choice off a complete ranking; the lower block observes it. Tail frequencies are floored at 1/201 by two hundred replicates. Forced-choice subgroups are quoted for the consumer experiments, since no deferral coding enters those.

3 Choices from real menus

Twelve of the thirteen collections used here are complete rankings, from which subset choice is read off rather than observed. One is not, and it comes first.

Costa-Gomes, Cueva, Gerasimou and Tejiščák [1] ran two experiments in which subjects chose separately from every non-singleton subset of five consumer goods, twenty-six menus; the first also included the five singletons. The archive names the goods by two-letter codes and we describe them no more precisely. Restriction is observed rather than imputed, which makes this the cleanest test in our collection, and the two experiments are separate studies with disjoint subjects, so we report them separately rather than pooled.

Subjects are split into five folds. On the training subjects, shares come only from the full five-item menu; both maps are calibrated to those shares and nothing else; each then predicts every menu of size two, three and four; and we score the choices actually made by held-out subjects. Intervals resample subjects and repeat the whole procedure, calibration included. Subjects who may defer are reported alongside the forced-choice subgroup, for whom no deferral coding enters. Of the 373 subjects, 364 record at least one usable choice from a non-singleton menu and enter the analysis.

The race is ahead in both experiments and in both treatments, and by more where deferral cannot interfere. The two experiments do not agree on strength. Experiment 1 clears its null comfortably, with Monte Carlo tail frequencies of 0.010 and 0.005; experiment 2, at fifty-eight per cent of the sample, does not, at 0.199 and 0.129. The direction replicates and the magnitude is not established by these data alone.

	n	Choices	Renorm.	Race	Gain	Excess
Experiment 1, all	230	5,134	0.9438	0.9298	+0.0140	+0.0158
Experiment 1, forced	76	1,900	0.9639	0.9197	+0.0442	+0.0466
Experiment 2, all	134	3,105	0.9162	0.9103	+0.0059	+0.0056
Experiment 2, forced	61	1,525	0.8939	0.8799	+0.0140	+0.0131
Pooled	364	8,239	0.9263	0.9164	+0.0100	+0.0123

Table 2: Held-out log loss per choice on menus of size two to four, both maps calibrated only to training-fold full-menu shares under a common add-α convention, $\alpha = 1/2$. Excess is the gain net of the median gain under a population that renormalizes exactly, as in Section 7. Subject-bootstrap intervals for the gains are $[+0.0072, +0.0335]$, $[+0.0171, +0.0867]$, $[+0.0009, +0.0306]$, $[+0.0034, +0.0560]$ and $[+0.0048, +0.0246]$ respectively.

4 Three more observed restrictions

The consumer experiment is not the only place restriction is observed. Three further populations are scored by the same protocol, and they differ in what does the removing.

A fourth was tried and discarded. Grocery stockout data records hourly sales by product, and a shopper may buy two items from one category, so the alternatives are not mutually exclusive and the object is a basket rather than a choice. Recovering choices would need transaction-level baskets, which the release does not contain.

Aguiar, Boccardi, Kashaev and Kim [9] observe choices from all thirty-one non-empty subsets of five lotteries with a default always available, so menus run from two to six alternatives, in a cross-section giving at most two disjoint menus per person. It is adversarial on the authors' own analysis, since they report that random utility cannot explain the population behaviour while a logit attention model survives.

The experiment was run three times on disjoint subject pools, differing only in how much arithmetic is needed to value an alternative: one operation, three, or five. The excess over the null rises with that requirement, from -0.0012 to $+0.0045$ to $+0.0081$, and the full-menu shares flatten alongside it, the largest falling from 0.357 to 0.257 against a six-alternative maximum entropy of 1.792.

We read this as the thesis working rather than as instability. When the arithmetic is one step the task has a computable answer and is closer to an examination question than to a preference judgement; subjects converge, the shares concentrate, and there is little preference left for any restriction map to describe. When it takes five steps nobody can compute the answer, so people fall back on what they actually prefer and the choice becomes psychological. A model of preference under noise should win in the second case and not the first. Winning in both, including where the answer is objectively determined, would be the result to distrust.

US presidential ballots supply restriction by statute. The alternatives are the same named candidates in every state and which of them appears varies with petition thresholds and filing deadlines, so the states are nested menus over identical alternatives. States are not interchangeable populations, so a state may only predict another whose two-party log odds are within 0.5.

Wikipedia links supply editorial removal. An article offers outbound links, a reader picks one exclusively, the destinations are the same articles from month to month, and editors change the set. Adjacent monthly clickstream dumps give the earlier month's shares and the later month's choices. The dump censors pairs below ten clicks, so a vanished destination may have been removed or may have fallen below threshold, and it records no link position, so layout is uncontrolled.

Population	Units	Gain	Null median	Excess	MC tail
Lotteries, high cost	4,099	+0.0101	+0.0020	+0.0081	.035
Lotteries, medium cost	4,099	+0.0068	+0.0023	+0.0045	.124
Lotteries, low cost	4,099	+0.0002	+0.0014	−0.0012	.667
US state ballots 2024	263 pairs	+0.0002	−0.0000	+0.0002	.005
Wikipedia links	3,544 pages	−0.0001	−0.0002	+0.0001	.005

Table 3: Held-out loss differences on populations where restriction is observed rather than read off a ranking. Excess is the gain net of what a renormalizing population produces on the same menus. The lottery frames are disjoint subject pools; ballot units are matched state pairs; Wikipedia units are pages whose destination set shrank; Two rows have a negative raw gain and a positive excess, which means the race loses to renormalization in absolute terms while still beating what the axiom predicts of it.

Two features deserve naming. The lottery gradient runs with arithmetic cost, and the mechanism of Appendix A predicts its direction, since computation error is per-alternative and roughly Gaussian, so loading more of it onto each alternative raises σ against β. And the Wikipedia rows say something the ranking collections cannot: tightening the filter that identifies a genuine removal from a five per cent share to thirty-five per cent raises the excess from +0.0001 to +0.0005 across 3,544, 817 and 121 pages, so the near-zero headline is attenuation by measurement rather than absence of an effect. Both Wikipedia and the two General Social Survey collections put the truth between the two defaults: choice contracts when an alternative goes, but by less than Case V says.

5 Choices induced from rankings

Twelve archival collections have human respondents and rankings, or ratings fine enough to yield them. Two rank-order card tasks from the General Social Survey ask what a child should learn and what matters in a job. Three independent samples of Jester users rate the same ten jokes. Sushi rankings come from a Japanese web survey, film rankings are induced from Netflix ratings, and dots and puzzles are perceptual discrimination tasks. The rest are rankings of seven sports by participation preference, ten occupations by perceived prestige, and four political goals from a five-nation survey. The thirteen collections comprise 147,719 responses, of which the menu experiment contributes 373 subjects; the twelve ranking collections contribute 147,346. These are not thirteen independent experiments, since three are Jester files, two are General Social Survey items, and dots and puzzles are companion tasks.

The protocol matches Section 3 with respondents in place of subjects, and the outcome for a subset is the respondent's highest-ranked member of it.

Averaging over every subset weights cardinalities by how many there are, so intermediate menus dominate: for $K = 10$ only 45 of 1,013 subsets are pairs, about four per cent of the weight. Those middling menus are where least has been removed and least is at stake. Splitting by menu size shows the pairwise gain exceeding the aggregate in eleven of twelve datasets, by factors from 0.7 to 3.7 with median 1.8: Sushi is +0.0412 at $|T| = 2$ against +0.0111 aggregate,

Dataset	n	K	Subsets	Renorm.	Race	Gain
Occupational prestige	143	10	1,013	1.0173	0.9783	+0.0390 [+0.0227, +0.0555]
Sushi	5,000	10	1,013	1.3724	1.3613	+0.0111 [+0.0094, +0.0128]
Political goals	2,262	4	11	0.8835	0.8766	+0.0069 [+0.0059, +0.0078]
GSS socialization	5,000	5	26	0.7984	0.7929	+0.0055 [+0.0041, +0.0070]
Jester file 3	5,000	10	1,013	1.5189	1.5165	+0.0024 [+0.0015, +0.0032]
Jester file 1	5,000	10	1,013	1.5287	1.5267	+0.0020 [+0.0012, +0.0027]
Jester file 2	5,000	10	1,013	1.5247	1.5230	+0.0018 [+0.0010, +0.0025]
GSS job values	5,000	5	26	0.8593	0.8576	+0.0017 [+0.0005, +0.0030]
Dots	3,183	4	11	0.8285	0.8270	+0.0015 [+0.0005, +0.0025]
Netflix	4,379	3	4	0.7932	0.7922	+0.0009 [+0.0007, +0.0012]
Puzzles	3,180	4	11	0.8180	0.8173	+0.0007 [−0.0003, +0.0017]
Sports participation	130	7	120	1.2579	1.2528	+0.0050 [+0.0017, +0.0087]

Table 4: Held-out log loss per prediction over every subset of size two or more. Datasets are capped at five thousand respondents for runtime. The race is lower in twelve of twelve and the interval excludes zero in eleven. Occupational prestige is scorable here only because the shared smoothing removes a zero training cell. Sports participation sits below the rule: its gain is real but Section 7 shows it carries no evidence about the restriction map. Intervals resample respondent losses with the fitted training models held fixed, so unlike Table 2 they omit calibration uncertainty and are too narrow, most seriously in the two smallest rows.

and GSS socialization +0.0130 against +0.0055. Occupational prestige is the exception, peaking at $|T| = 4$. At $|T| = K$ nothing has been removed, the two rules must agree, and the measured gain is zero to four decimals in every dataset, which is the check that the pipeline is sound. We keep the all-subsets aggregate as the headline because it fixes the estimand in advance rather than selecting a cardinality after the fact.

6 Where the difference lives

Averaged over outcomes the difference between the two defaults is small, and averaging hides what it is made of. Two decompositions locate it.

Score each held-out prediction by the rank the chosen item held in the full-menu shares. Pooling the four ten-item collections:

Rank of chosen item	1	2	3	4	5	6	7	8	9	10
Share of outcomes	.17	.13	.13	.10	.10	.10	.08	.07	.07	.04
Mean gain	−.062	−.045	−.034	−.004	+.009	+.019	+.050	+.058	+.101	+.171

Table 5: Held-out gain by the full-menu rank of the item actually chosen, pooled over Sushi and the three Jester files. Monotone in rank, crossing zero between fourth and fifth. The contributions sum to +0.0043.

The race pays 0.062 nats whenever the favourite wins, which happens 17 per cent of the time, and collects 0.171 when the tenth-ranked item wins, which happens 4 per cent of the time. That is what a contraction does: it moves probability off the favourite and onto the tail, buying insurance against upsets and paying a premium on every favourite win. Three earlier observations follow from it. The aggregate gain is small because large rare wins net against small frequent losses. The ballot result is negligible because third-party candidates are the tail and there is almost no tail mass to collect. And grocery baskets are a wash because a category where both items can be bought has no upsets.

The second decomposition is sharper because it names an error rather than a difference. Ask how often a given item finishes second, which is the conditional question after the winner is removed:

Collection	Observed	Renormalization	Race
Political goals	0.123	0.318	0.309
GSS job values	0.186	0.303	0.294
GSS socialization	0.191	0.302	0.294
Sushi	0.206	0.251	0.245
Jester file 1	0.138	0.160	0.158

Table 6: Probability that the full-field favourite finishes second. Renormalization overstates it in every collection, by 0.022 to 0.195; the race overstates it too, by a little less. The mirror holds for the last-ranked item, whose chance of finishing second both maps understate everywhere.

This is the pattern reported for the Harville formula in racing place markets, favourites over-priced and outsiders under-priced, appearing here in survey rankings of sushi, jokes, child-rearing values and political goals. It is also the most useful result in the paper for a practitioner, because it is a large, replicable, signed error in the default rather than a small average edge. The honest qualification is that the race removes only 5 to 12 per cent of it, so both defaults share a bias much larger than the distance between them. What produces the remainder is not identified here; the obvious candidate is that a single respondent's ranking is internally consistent, so adjacent places are positively correlated in a way no independent contest reproduces.

7 Shrinkage

A positive gain has two possible sources and log loss cannot tell them apart. The race may be closer to how the population chooses. Or the race may be wrong about that and win anyway, because contracting a noisily estimated share vector is shrinkage, and shrinkage lowers log loss when the input is over-confident. The second mechanism needs no departure from the axiom, so properness of the score licenses nothing here: a proper score says the true probability vector is optimal, not that a plug-in estimate of it beats a biased transformation of the same estimate.

The remedy is to build a population where the axiom holds by construction and run the identical procedure on it. Taking the observed full-menu shares as worths, choices or complete rankings are generated from that exact process, and everything is repeated. Rankings are drawn by sorting $\log u_i + G_i$ for independent standard Gumbel G_i, which is the Plackett–Luce ranking law; for $K \geq 4$ this is a particular joint law consistent with those subset marginals rather than the only one, so the null is specifically a Plackett–Luce ranking null.

The shrinkage mechanism is real and it accounts for the entire gain in the smallest dataset. Sports participation, with 130 respondents, sits exactly on its null and is excluded from every claim here. Everywhere else the mechanism runs against the race, and the excess over the null exceeds the raw gain in all twelve remaining rows. It also rescues puzzles, whose raw interval in Table 4 covers zero while its excess over the null does not. The menu experiment clears the null

Dataset	Observed	Null median	Excess	MC tail
Menu experiment, forced choice	+0.0265	−0.0041	+0.0306	0.010
Menu experiment, all subjects	+0.0100	−0.0023	+0.0123	0.015
Occupational prestige	+0.0390	−0.0227	+0.0617	$\leq$ 0.005
Sushi	+0.0111	−0.0062	+0.0174	$\leq$ 0.005
GSS socialization	+0.0055	−0.0066	+0.0121	$\leq$ 0.005
Political goals	+0.0069	−0.0012	+0.0081	$\leq$ 0.005
GSS job values	+0.0017	−0.0047	+0.0065	$\leq$ 0.005
Jester file 3	+0.0024	−0.0015	+0.0039	$\leq$ 0.005
Jester file 1	+0.0020	−0.0014	+0.0034	$\leq$ 0.005
Jester file 2	+0.0018	−0.0014	+0.0032	$\leq$ 0.005
Dots	+0.0015	−0.0014	+0.0029	$\leq$ 0.005
Puzzles	+0.0007	−0.0011	+0.0019	$\leq$ 0.005
Netflix	+0.0009	−0.0001	+0.0010	$\leq$ 0.005
Sports participation	+0.0050	+0.0051	−0.0000	0.51

Table 7: The same gain when the axiom holds by construction. Two hundred replicates per row. The diagnostic is the excess of the observed gain over the null median; the tail probabilities are indicative rather than exact tests, and for the menu rows they are optimistic, because the null redraws each subject's thirty-one choices independently given the aggregate worths and so omits the within-subject dependence that the bootstrap intervals of Table 2 do preserve. Wherever shares rest on more than about two thousand observations the null is negative, because contraction is then a bias rather than useful insurance, so comparing an observed gain with zero understates it.

on both readings, which is the result the paper rests on, since it is the only one where restriction is observed.

8 The benchmark gap

Most of the aggregate gains in Tables 2 and 4 are a few thousandths of a nat, and such a figure is hard to read on its own. Neither map is permitted to see a restricted menu, and most of what is predictable about a restricted menu is already contained in the full-menu probabilities that both of them receive.

We therefore score a third forecast which is permitted to see them. Fix a subset T and a training fold. Among the training respondents, count how often each item of T was the one taken from T, smooth those counts as above, and use them as the forecast for held-out respondents facing T. This is a direct estimate of the choice probabilities within T, built from exactly the observations the two maps are denied, and it is scored on the same held-out respondents by the same log loss.

It beats both maps, as it should. The question is by how much, because that difference is the room available. Define the gap as the held-out loss reduction of this counting forecast relative to renormalization, and read the race's share of that gap from Table 8.

That third forecast is not an oracle. It is estimated too, so its expected held-out loss exceeds the true conditional entropy by approximately $(m-1)/2n$ for an m-outcome menu and n training observations. The last two columns add that correction to the gap, which lowers the median

Dataset	Gap	Captured	Corrected gap	Captured
Dots	0.0023	65%	0.0026	58%
Jester file 1	0.0041	49%	0.0046	43%
Jester file 3	0.0051	47%	0.0056	43%
Political goals	0.0150	46%	0.0154	45%
Jester file 2	0.0040	45%	0.0045	40%
Sushi	0.0260	43%	0.0265	42%
Occupational prestige	0.1135	34%	0.1312	30%
Puzzles	0.0026	27%	0.0029	24%
GSS socialization	0.0223	25%	0.0225	24%
Netflix	0.0046	20%	0.0048	19%
GSS job values	0.0087	20%	0.0089	19%
Median		43%		40%

Table 8: Gap is the held-out loss reduction of a smoothed per-subset frequency forecast relative to renormalization. Captured is the race's share of it. The eleven are the ranking collections whose observed gain exceeds the corresponding null simulation in Table 7; sports participation is excluded throughout. Renormalization captures none of the gap, being the unique map that preserves every ratio.

from 43 to 40 per cent.

Two of the gaps are near 0.002, so the dots and puzzles ratios are the least stable. The percentages are context for the absolute gains.

9 Scope

The practical recommendation is narrow. When the task is to infer restricted-menu population shares from full-menu population shares, the Gaussian race should be the default in place of renormalization. The inversion from shares to locations is the only extra machinery, and the lattice algorithm of [2] performs it in one pass over a common grid, returning the whole vector of winning probabilities at once. The restricted predictions are then one-dimensional quadrature on the same grid. At the field sizes here, ten alternatives at most, both steps are immediate; a field of $n = 10{,}000$ takes a fraction of a second, and the method scales to fields of millions [2]. So the recommendation needs no additional data and no appreciable additional computation, and it was better on every collection examined here except the one too small to discriminate. In the single experiment where restriction is actually observed rather than induced from rankings, it is better by $+0.0265$ nats per choice among subjects who cannot defer.

All twelve ranking collections carry an assumption the menu experiments do not. A respondent's ranking imposes one ordering on every subset by construction, so no ranking dataset can show a person choosing i over j from one menu and j over i from another, and none reveals whether that person would choose as their ranking implies when handed two options. Reading restriction off a ranking assumes they would. Section 3 does not.

Every quantity here is a population quantity. A population of people who each choose by Equation (1), with worths differing between people, generally does not satisfy the axiom in aggregate, which is why mixed logit is useful and why its probabilities are integrals over logit probabilities rather than logit probabilities [7]. Aggregate failure of the axiom is therefore not evidence that individuals violate it. Heterogeneous Luce choosers can reproduce aggregate contraction, and Appendix A exhibits a class in which they do. What is rejected is a representative-agent map.

The experiment compares two restriction maps, not two fitted likelihoods. A fitted Plackett–Luce model [14] uses the whole ranking and a fitted Bradley–Terry model [10] uses the pairwise outcomes directly. Neither was estimated here. Recommendations about ranking or paired-comparison likelihoods therefore have no support in these tables.

The score is a sum of marginal scores for top-item-within-subset outcomes. It is proper for those marginals and not for the ranking distribution. With $K = 4$ the scored marginals carry at most 17 degrees of freedom against 23 for a distribution over rankings, so distinct ranking laws can agree on everything scored here.

Nesting is not costlessness. Conceptually the independent-noise race family nests Luce at the Gumbel noise law, so the family excludes no Luce population, but Table 7 shows that choosing the Gaussian point is a separate statistical decision which loses when the axiom holds and the shares are well estimated.

So the recommendation is not a claim of dominance. Neither map dominates the other: when the shares are precise the better map is whichever one matches the population, and when they are very noisy the dominant consideration is estimation risk rather than either structural rule. What the tables support is a structural default in the absence of evidence for the axiom, which is the situation an analyst holding only aggregate shares is actually in.

A tempting shortcut fails. Pricing ordered outcomes by removing the winner and re-running looks like the race's own prediction, and it is not: against the calibrated race's ordering law it misprices individual pairs and triples by up to a factor of 3.4. We leave to future work the comparison of the exact Gaussian ordering law against Plackett–Luce ranking probabilities on complete-ranking data.

Data and code

Sushi rankings come from Kamishima [4]; Jester ratings from the Eigentaste project [3]; the General Social Survey cumulative file from NORC; the perceptual, leisure, prestige, film and political-goal rankings via PrefLib [6] and the CRAN packages PLMIX and ConsRank; the menu-choice data from the replication archive of [1]. All are public. Analysis code and every input needed to reproduce the tables are under `research/human` on the `machine-preference-paradox` branch of `github.com/microprediction/winning`, the inputs as column extracts rather than raw archives.

Two storage conventions must be told apart when reproducing this. BayesMallows, ConsRank and PerMallows store ranks, so element j holds the rank of alternative j, while PLMIX stores orderings, so the first element names the favourite. Reading one as the other transposes the data silently and raises no error. We established which is which by cross-validation: the political goals data ships in two of these packages and they agree only when each is read under its own convention.

A Why a population should look Gaussian

Nothing so far explains why the Gaussian member of the class should be the one that fits. A mechanism is available, and it is elementary.

Let every individual obey the axiom, and let their tastes differ. Individual θ assigns systematic utility $V_i(\theta)$ to alternative i and chooses by Luce at scale $\beta > 0$,

$$P(i \mid S, \theta) = \frac{\exp\{V_i(\theta)/\beta\}}{\sum_{j \in S} \exp\{V_j(\theta)/\beta\}}, \tag{4}$$

which by Yellott's theorem is the same as choosing the largest $V_i(\theta) + \beta G_i(\theta)$ for independent standard Gumbel $G_i(\theta)$.

Proposition 2 (Gaussian taste heterogeneity)**.** *Suppose $V_i(\theta) = \mu_i + \sigma Z_i(\theta)$ with $Z(\theta) \sim N(0, I_K)$ independently across individuals. Then the aggregate choice probabilities are exactly the winning probabilities of a contest with utilities*

$$\mu_i + \sigma Z_i + \beta G_i, \tag{5}$$

independent across alternatives. If $\mu_i = \sigma a_i$ and $\sigma/\beta \to \infty$, they converge to those of Case V with locations a.

Proof. Aggregating over individuals integrates over the taste draw Z, leaving the per-alternative utility (5), independent across i because both components are. Dividing every utility by σ preserves the $\arg\max$ and gives $a_i + Z_i + (\beta/\sigma)G_i$, whose last term vanishes in probability. Ties have probability zero under the Gaussian limit, so the winning probabilities converge. □

This is a mixed logit: Gaussian heterogeneity between individuals, Gumbel noise within them. Such models are standard, and McFadden and Train [7] establish how general they are. Nothing here claims otherwise. What the proposition supplies is the connection between the high-heterogeneity limit of that familiar object and the particular restriction map the paper tests.

The Gaussian assumption on taste is where a limit theorem belongs.

Corollary 1 (Many small taste components)**.** *Suppose $V_i(\theta) = \sqrt{m}\, a_i + \sum_{\ell=1}^{m} \xi_{\ell i}(\theta)$, where the vectors $\xi_\ell(\theta)$ are independent and identically distributed across ℓ with mean zero, covariance I_K and finite second moments, and the individual chooses by (4) at fixed β. Then the aggregate choice probabilities converge as $m \to \infty$ to those of Case V with locations a.*

Proof. Dividing the utilities by $\sqrt{m}$ gives $a_i + m^{-1/2} \sum_\ell \xi_{\ell i} + (\beta/\sqrt{m})G_i$. The multivariate central limit theorem sends the middle term to $N(0, I_K)$ and the last to zero, and the $\arg\max$ probabilities are continuous at a limit with no ties. □

One direction for future work would turn this limit into a correction. At finite σ/β the composite noise is $Z + \varepsilon G$ with $\varepsilon = \beta/\sigma$, so the natural device is a cumulant expansion about the Gaussian rather than anything singular. The first two Gumbel cumulants do little: the mean is a common location shift and cancels in the $\arg\max$, and the variance inflation $1 + \pi^2\varepsilon^2/6$ is a common scale change, which would be absorbed entirely if the calibration matched the whole law. It does not. It matches the $K-1$ full-menu shares, so a common scale error is invisible in the fit and propagates into the restricted-menu prediction, which is the quantity at issue. Simulating the deviation of observed contraction from the recalibrated Case V value over $\varepsilon \in [0.125, 1]$ gives local exponents of 1.6, 2.0 and 2.3, so the range we can resolve looks quadratic rather than cubic; the smallest point sits near Monte Carlo resolution and deterministic quadrature would be needed to settle the rate. Either way the target is already tabulated, since the ratio column above is what such an expansion must reproduce.

The covariance assumption is doing real work and should not be waved through. If the taste components carry covariance Σ instead of I_K, the same argument delivers a Gaussian race with covariance Σ, which is correlated multinomial probit rather than Case V. Case V requires the covariance relevant to utility differences to be spherical. So the limit theorem motivates Gaussian random utility in general, and independence across alternatives is a further restriction, adopted here because it is the parameter-free member of the class.

σ	Top share	Observed λ	Case V λ	Ratio
0.5	0.255	0.052	0.225	0.23
1	0.300	0.128	0.227	0.57
2	0.343	0.197	0.229	0.86
3	0.359	0.216	0.229	0.94
4	0.365	0.228	0.230	0.99
8	0.373	0.230	0.231	1.00
16	0.375	0.233	0.231	1.01

Table 9: A population of Luce choosers whose log-worths carry Gaussian taste variation of scale σ against a Gumbel choice-noise scale of 1, with locations scaled as σa so that the shares do not degenerate. Observed λ is the population's contraction; Case V λ is what the independent unit-variance Gaussian race implies for that population's own shares. The top share is reported to show the limit is not the near-indifference corner.

The consequence for this paper is that no individual need be non-Lucean for the Gaussian transport to be the right one upstairs. A population of Luce choosers whose tastes vary is a contest with composite noise, and as taste variation comes to outweigh within-person choice noise the composite approaches Gaussian and the population contraction approaches the Case V prediction. That mechanism also says where the fit should be worst: in populations whose members are nearly alike.

Independent evidence for the same phenomenon exists in memory research. Robinson and colleagues [8] compare softmax against signal detection across m-alternative tasks and report that the softmax parameter must vary with m while the signal-detection parameter is stable. That is the contraction studied here, seen through the parameter that has to move to absorb it.

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